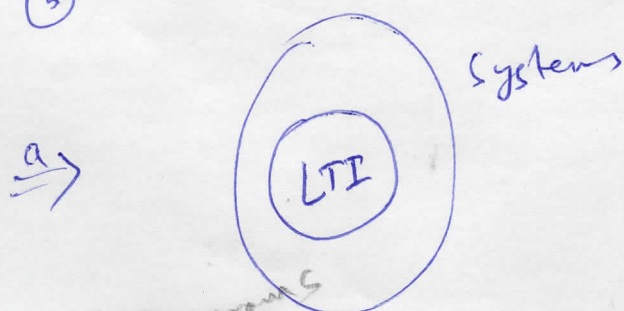


Q. How to find the output of an arbitrary system to an arbitrary input?

Problem $\begin{cases} \text{a) So Many types of systems!} \\ \text{b) So Many signals! (input candidates)} \end{cases}$

$\begin{cases} \text{a) Perhaps consider first a subclass of systems?} \\ \text{b) Perhaps consider a simple model that can cover all signals?} \end{cases}$



see also diagrams
block on next
page

Linear

In other words $\Rightarrow$ for linear system if $S(x_1[n]) = y_1[n]$
and $S(x_2[n]) = y_2[n]$ then $S(ax_1[n] + bx_2[n]) = aS(x_1[n]) + bS(x_2[n]) = ay_1[n] + by_2[n]$

TI



In other words: for TI system if $S(x[n]) = y[n]$
then $S(x[n-N]) = y[n-N]$

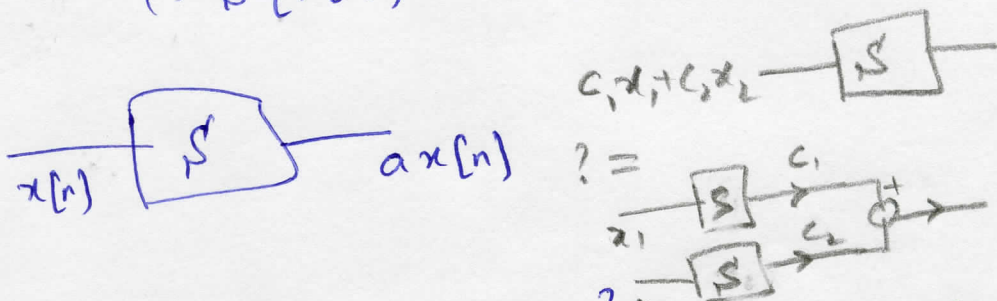
e.g are these LTI?

(i) $S[x(n)] = ax(n)$

(ii) $S[x(n)] = ax(n) + b$

(iii) $S[x(n)] = ax(n) + bn$

Let's see (i)



Linearity test

$$S[c_1 x_1(n) + c_2 x_2(n)] \stackrel{?}{=} c_1 S[x_1(n)] + c_2 S[x_2(n)]$$



$$\stackrel{?}{=} S[x(n)] \rightarrow D^n \rightarrow a c_1 x_1(n) + a c_2 x_2(n)$$

$$\stackrel{\checkmark}{=} c_1 a x_1(n) + c_2 a x_2(n)$$

TI test

$$S[x(n-N)] \stackrel{?}{=} y[n-N]$$

input delayed output delay

with $y[n] = S[x(n)]$
with $y[n] = ax(n)$

$$a x(n-N) \stackrel{\checkmark}{=} a x(n-N)$$

(ii) $S[c_1 x_1(n) + c_2 x_2(n)] \stackrel{?}{=} c_1 S[x_1(n)] + c_2 S[x_2(n)]$

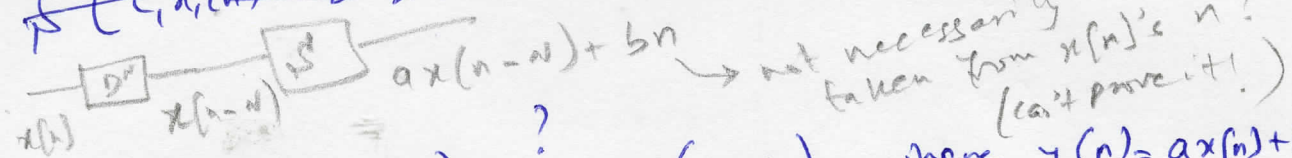
not linear

$$a c_1 x_1(n) + a c_2 x_2(n) + b \stackrel{x}{=} c_1 a x_1(n) + c_1 b + c_2 a x_2(n) + c_2 b$$

TI $\checkmark$ $S[x(n-N)] \stackrel{?}{=} y[n-N] \rightarrow$ with $y[n] = ax(n) + b$

$$a x(n-N) + b \stackrel{\checkmark}{=} a x(n-N) + b$$

(iii) $\mathcal{S}(c_1 x_1(n) + c_2 x_2(n))$ not linear (easy to see)



not necessarily taken from $x(n)$'s n ! (can't prove it!)

not
TI

$$\mathcal{S}(x(n-n)) = y(n-n) \rightarrow \text{where } y(n) = ax(n) + bn$$

$$ax(n-n) + \underline{bn} = ax(n-n) + b(n-n)$$

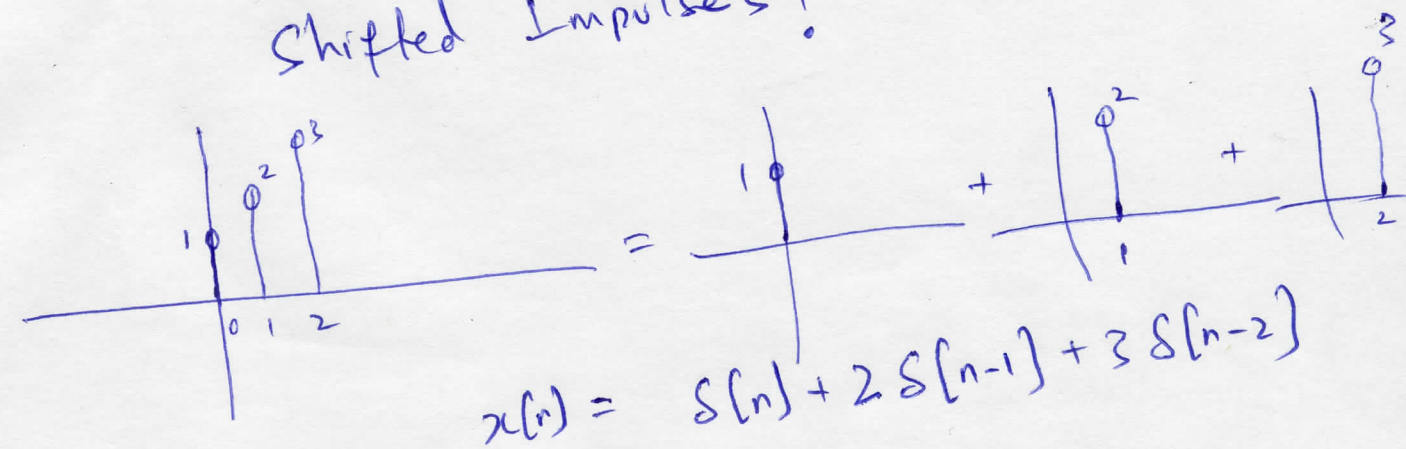
not linked to input, so won't change

→ o.k, so let's stick to LTI for convenience

→ now let's find a generic signal model.

claim Any DT signal can be written as a

Sum of
Scaled and
Shifted Impulses!

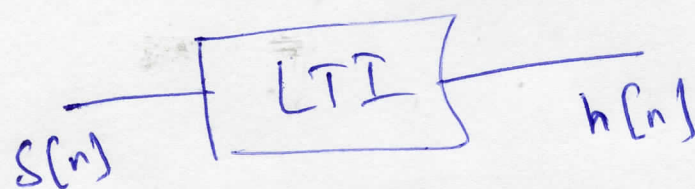


$$x[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2]$$

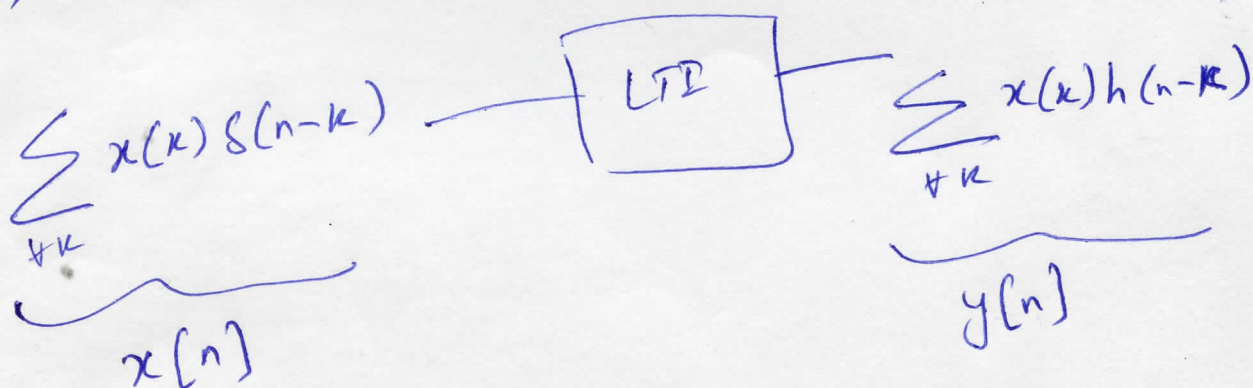
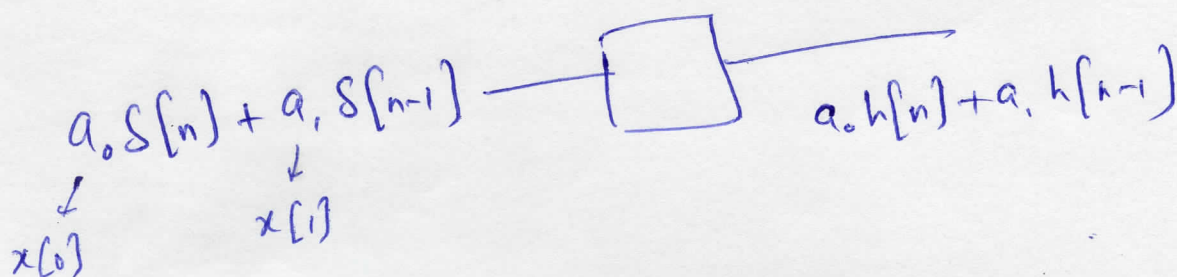
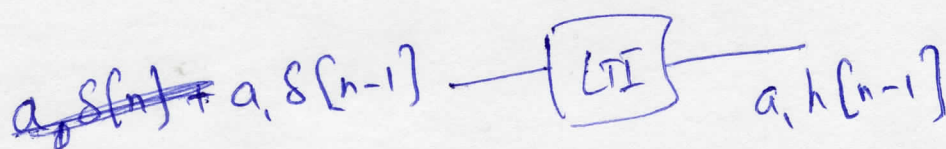
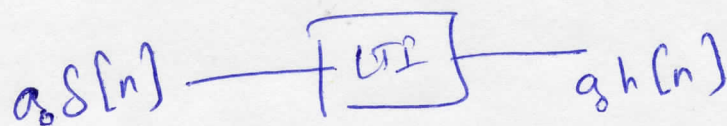
$$x[n] = \sum_k x[k] \delta[n-k]$$

→ now let's bring these two together!

(4)



let's call it
"Impulse
Response".



$$y[n] = \sum_{k=-K} x(k) h[n-k]$$

$$= x[n] * h[n]$$

$$= \underline{\underline{\text{Convolution}}}$$

So basically

→ Convolution gives us the output of an LTI system to an arbitrary signal! (you need to know only the input signal and IR)

→ So two take aways

Convolution $\Leftarrow$ (mathematical operator 5)
 helps find LTI response

IF $(h(n)) \leftarrow$ can fully characterize an LTI system
 (as we shall see soon)

→ both of these need full attention!

useful
 → convolution properties

identity

$$x[n] * \delta[n] = x[n]$$

Commutative

$$x_1[n] * x_2[n] = x_2[n] * x_1[n]$$

$$\stackrel{\text{so}}{=} \sum_{k} x[k] h[n-k] = \sum_{k} h[k] x[n-k]$$

Distributive

$$x_1[n] * (x_2[n] + x_3[n]) = (x_1[n] * x_2[n]) + (x_1[n] * x_3[n])$$

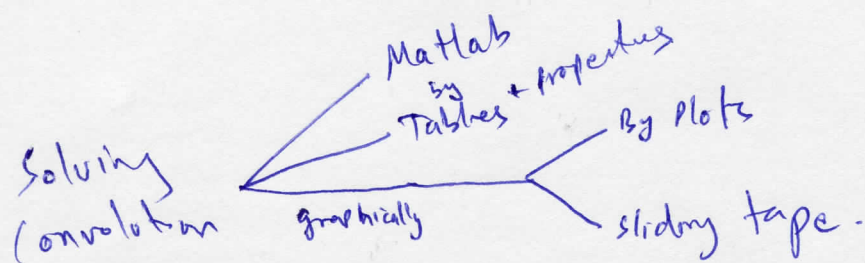
Associative

$$x_1[n] * (x_2[n] * x_3[n]) = (x_1[n] * x_2[n]) * x_3[n]$$

Shifting if $x_1[n] * x_2[n] = c[n]$
 then $x_1[n-m] * x_2[n-p] = c[n-m-p]$

⑥

→ like Integrals, we have convolution tables.
 → also mostly done on computers.



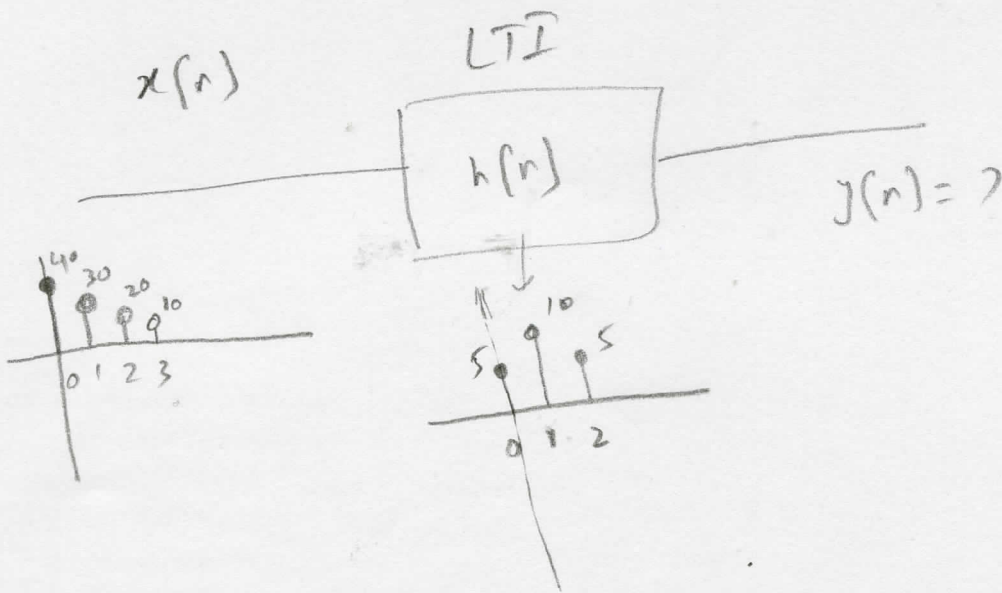
① Matlab $x = [-1 \ 0 \ 1]$, $h = [0 \ 1 \ 0]$
 $y = \text{conv}(x, h)$ ← command.

② $u[n] * (s[n] + su[n])$?

by Dist. prop. → $(u[n] * s[n]) + u[n] * su[n]$

by table → $u[n] + s[(n+1)u[n]]$

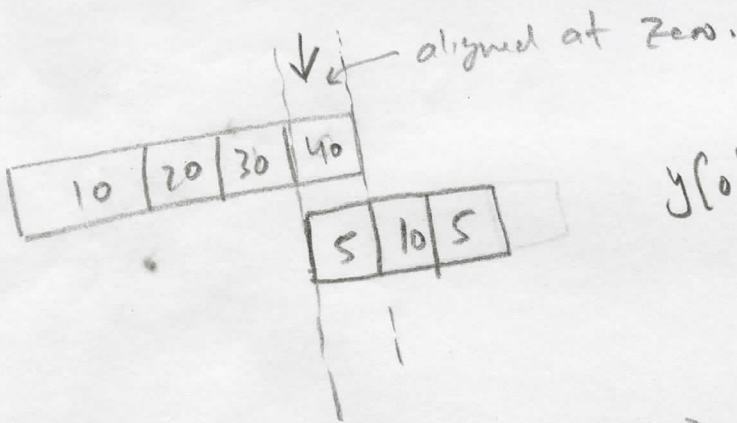
③ By plots & By tape similar (same basic idea)
 ↓
 easier to draw



$$y[n] = x[n] * h[n] = \sum_{k} x[k] h[n-k] = \sum_{k} h[k] x[n-k]$$

$$\Rightarrow y[0] = \sum_{k} h[k] x[-k] \leftarrow \text{input flipped}$$

$$\Rightarrow y[1] = \sum_{k} h[k] x[-k+1] \leftarrow \text{flipped \& shifted}$$



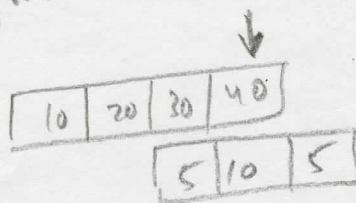
$$y[0] = 5 \times 40 = 200$$

Step 1 Flip the input (or IR)

Step 2 Align input & IR at zero.

Step 3 Multiply overlapping cells & add results

Step 4 Shift right & repeat.



$$y[1] = 30 \times 5 + 40 \times 10 = 150 + 400 = 550$$